IT114 Module 2 Java Challenge Scenarios

Student submission

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Worksheet link

https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-module-2-java-challenge-scenarios/aia47

Grading link

https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-module-2-java-challenge-scenarios/grading/aia47

1 submission warning(s) were acknowledged.

AI disclosure: Completed; 1 entry; 0 declared AI use; 1 declared no AI use; policy v1

Java Scenarios

Complete each starter scenario, prove its source and output separately, and explain the decisions used in the solution.

Scenario 1 source - odd values

Edit `printOdds` so each passed array prints only odd values on one line, separated by comma and space, with no leading or trailing comma.

URL prompt

Submit the GitHub URL that opens `M2/Scenario1.java` on your `M2-Homework` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- `printOdds` selects only odd values. (up to -0.35 points)
- `printOdds` places comma-space separators only between printed values. (up to -0.35 points)

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/M2-Homework/M2/Scenario1.java

Scenario 1 output

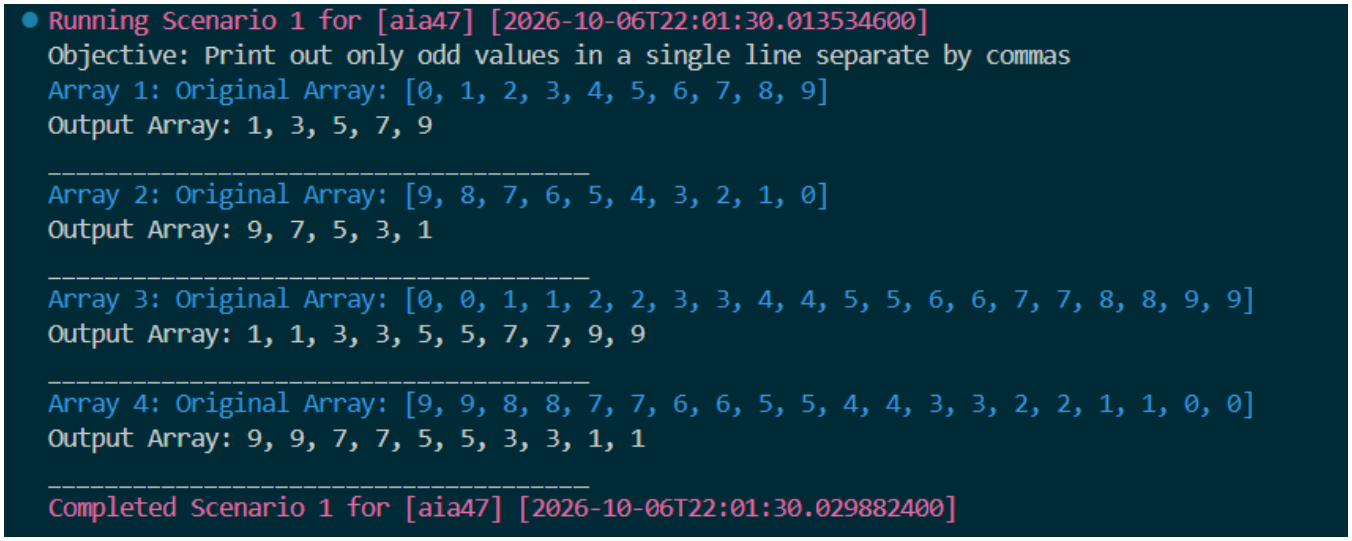
Image evidence prompt

Upload one readable image of the complete `Scenario1` program output produced from your `M2-Homework` branch.

Requirements and potential deductions

- The output contains only the expected odd array values. (up to -0.35 points)
- Comma-space separators occur only between printed values. (up to -0.2 points)

Caption: screenshot of output



Scenario 1 explanation

Response prompt

Explain how the code on `M2-Homework` detects odd values and avoids a leading or trailing separator.

Requirements and potential deductions

- The explanation identifies the odd-value test used in the linked code. (up to -0.35 points)
- The explanation identifies the separator decision used in the linked code. (up to -0.2 points)

Submitted response

There is a boolean to check if an odd value has been printed, that is the identifier for separator. For the odd-value test there is loop that goes through every value in the array and checks if the value is odd.

Scenario 2 source - sum and display

Edit `sumValues` to total the passed array in `total` and assign a representation with exactly two decimal places to `modifiedTotal`.

URL prompt

Submit the GitHub URL that opens `M2/Scenario2.java` on your `M2-Homework` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- `sumValues` stores the correct raw array sum in `total`. (up to -0.35 points)
- `modifiedTotal` represents the sum with exactly two decimal places. (up to -0.35 points)

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/M2-Homework/M2/Scenario2.java

Scenario 2 output

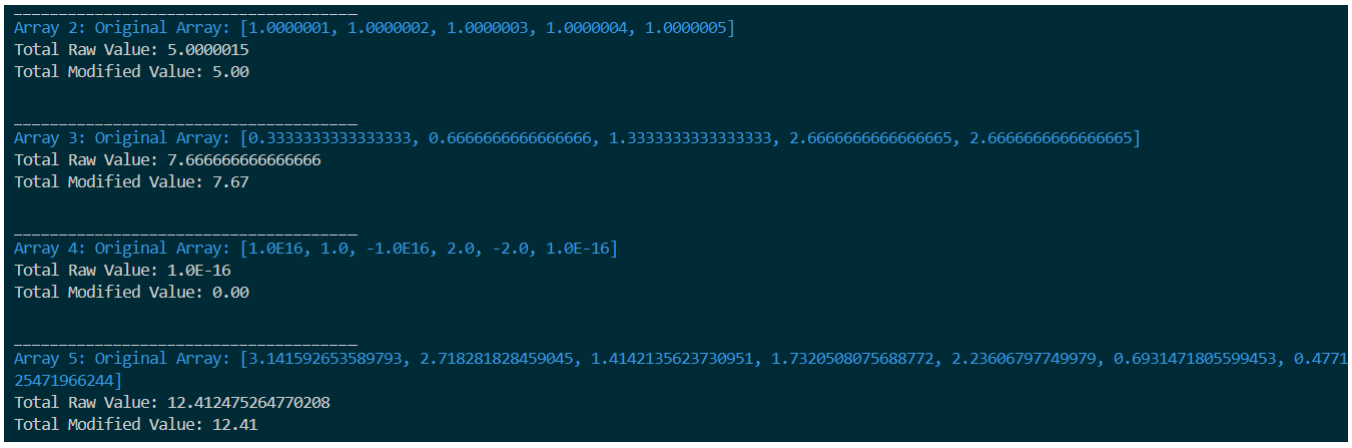
Image evidence prompt

Upload one readable image of the complete `Scenario2` program output produced from your `M2-Homework` branch.

Requirements and potential deductions

- The output shows the correct computed sums for the starter arrays. (up to -0.35 points)
- The display results have exactly two decimal places. (up to -0.2 points)

Caption: output for scenario 2



Scenario 2 explanation

Response prompt

Explain the `M2-Homework` solution's total and two-decimal representation. Independently check the math for the starter inputs, compare expected and observed results, and investigate the discrepancy and its likely cause.

Requirements and potential deductions

- The response explains the raw-sum calculation used in the linked code. (up to -0.25 points)

- The response explains how the display is represented with two decimal places. (up to -0.1 points)
- The response compares the observed arithmetic with an independently checked expected result and investigates the discrepancy and its likely cause. (up to -0.2 points)

Submitted response

There is a for loop that checks every value in the array and adds that value to the empty variable total. then empty object modifiedTotal prints a formatted string using String.format. The discrepancy is the answer for array 4,the output retruns 0 desp answer being 1. The most likely cause is a rounding error from formatting.

Scenario 3 source - positive conversion

Edit `bePositive` to make each value positive, convert it back to its original data type, and place it in the corresponding output array slot.

URL prompt

Submit the GitHub URL that opens `M2/Scenario3.java` on your `M2-Homework` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- `bePositive` makes each numeric value positive. (up to -0.25 points)
- Each converted result has its original data type. (up to -0.25 points)
- Each result is stored in the corresponding output array slot. (up to -0.2 points)

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/M2-Homework/M2/Scenario3.java

Scenario 3 output

Image evidence prompt

Upload one readable image of the complete `Scenario3` program output produced from your `M2-Homework` branch.

Requirements and potential deductions

- The output shows the expected positive results. (up to -0.35 points)
- Each displayed result corresponds to the original array position. (up to -0.2 points)

Caption: output for scenario 3

```
Output:
3.14159265358979[D], 2.718281828459[D], 1.61803398875[D], 0.5772156649[D], 1.0E-7[D], 1000000.0[D]

---------------------------------------
Array 3: Original Array:
1.1[F], -2.2[F], 3.3[F], -4.4[F], 5.5[F], -6.6[F], 7.7[F], -8.8[F]

Output:
1.1[F], 2.2[F], 3.3[F], 4.4[F], 5.5[F], 6.6[F], 7.7[F], 8.8[F]

---------------------------------------
Array 4: Original Array:
123[S], -456[S], 789.01[S], -234.56[S], 0.00001[S], -99999999[S]

Output:
123.0[S], 456.0[S], 789.01[S], 234.56[S], 1.0E-5[S], 9.9999999E7[S]

---------------------------------------
Array 5: Original Array:
-1[I], 1[I], 2.0[F], -2.0[D], 3[S], -3.0[S]

Output:
1[I], 1[I], 2.0[F], 2.0[D], 3.0[S], 3.0[S]
```

Scenario 3 explanation

Response prompt

Explain the conversion and index decisions in your `M2-Homework` solution.

Requirements and potential deductions

- The response explains how the linked code preserves the original type. (up to -0.2 points)
- The response explains how the linked code selects the corresponding output position. (up to -0.35 points)

Submitted response

Used instanceof to get class and used Math.abs to get absolute values alongside perserving types. The loop uses count i and taht into output i so that the values are pritned in the same place and type.

Scenario 4 source - text transformation

Edit `transformText` so the combined result removes non-alphanumeric characters except spaces, uses Title Case, removes leading and trailing whitespace, collapses duplicate interior spaces, and assigns the result to `placeholderForModifiedPhrase`.

URL prompt

Submit the GitHub URL that opens `M2/Scenario4.java` on your `M2-Homework` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- `transformText` removes non-alphanumeric characters except spaces. (up to -0.25 points)
- The source converts the phrase to Title Case. (up to -0.15 points)
- The source removes leading and trailing whitespace from the phrase. (up to -0.1 points)
- The source collapses duplicate interior spaces. (up to -0.1 points)
- The source assigns the final phrase to `placeholderForModifiedPhrase`. (up to -0.1 points)

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/M2-Homework/M2/Scenario4.java

Scenario 4 output

Image evidence prompt

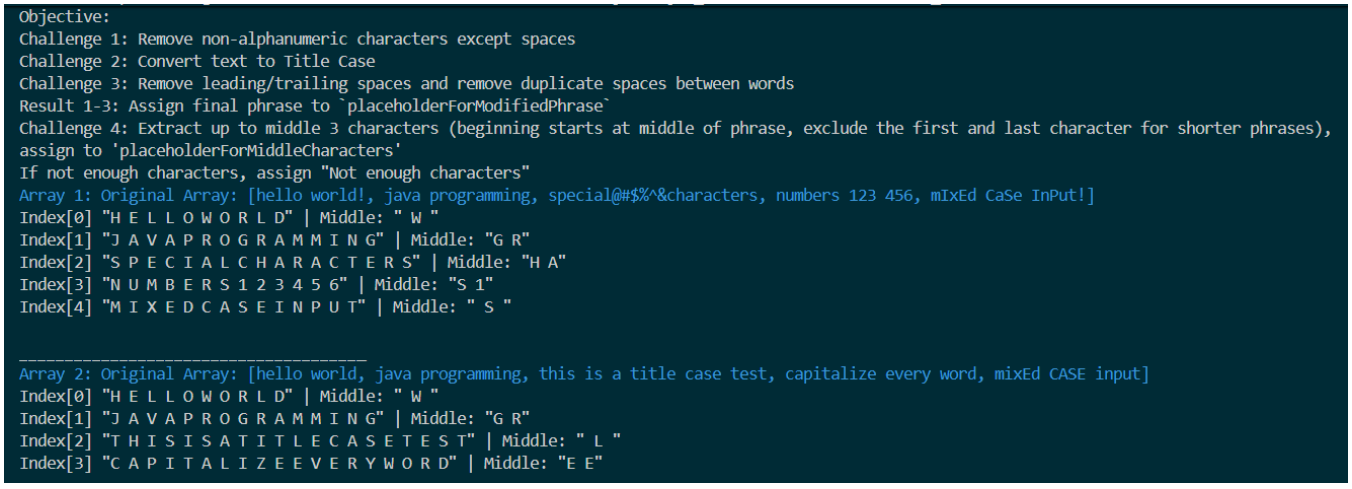
Upload one readable image of the complete `Scenario4` program output produced from your `M2-Homework` branch.

Requirements and potential deductions

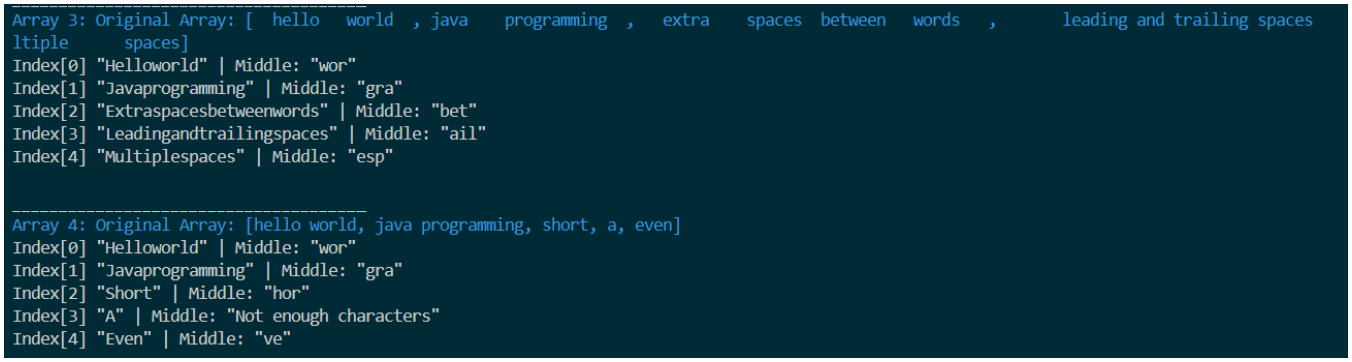
- The output phrases contain no disallowed non-alphanumeric characters. (up to -0.25 points)
- The output phrases use Title Case. (up to -0.1 points)
- The output phrases have no leading or trailing spaces. (up to -0.1 points)
- The output phrases have no duplicate interior spaces. (up to -0.1 points)

Warning at submission: expected-items

Caption: output for scenario 4 - Before I realized that there was an error in my code



Caption: output for scenario 4 - After I fixed code (view commit message for fix)



Scenario 4 explanation

Response prompt

Explain the order of transformations and how the `M2-Homework` solution finds word boundaries.

Requirements and potential deductions

- The response explains the sequence of text-processing operations in the linked code. (up to -0.35 points)
- The response explains how the linked code finds word boundaries. (up to -0.2 points)

Submitted response

Make a for loop that uses a local variable and indexing to remove banned characters to spaces. In same loop trim strings and c
string to lowercase so mixed casing doesn't matter. Boundaries are found through the split command " " to separate making ar
of words that are separated by 1 space when put back together.

Reflection

This section is worth one point total: submit your pull-request URL and separate WakaTime project/file evidence, then answer the three reflection questions.

Pull request and staged history

URL prompt

Submit the GitHub URL for the pull request from `M2-Homework` into `main`; its Commits tab must show the assignment's staged history.

Requirements and potential deductions

- The pull request targets `main` from `M2-Homework` in the personal repository. (up to -0.05 points)
- Its Commits tab shows the starter baseline, planning-comment commits made before solutions, and subsequent solution commits. (up to -0.15 points)

https://github.com/aia47-cyber/aia47-IT114-001-f2026/pull/3

WakaTime project activity

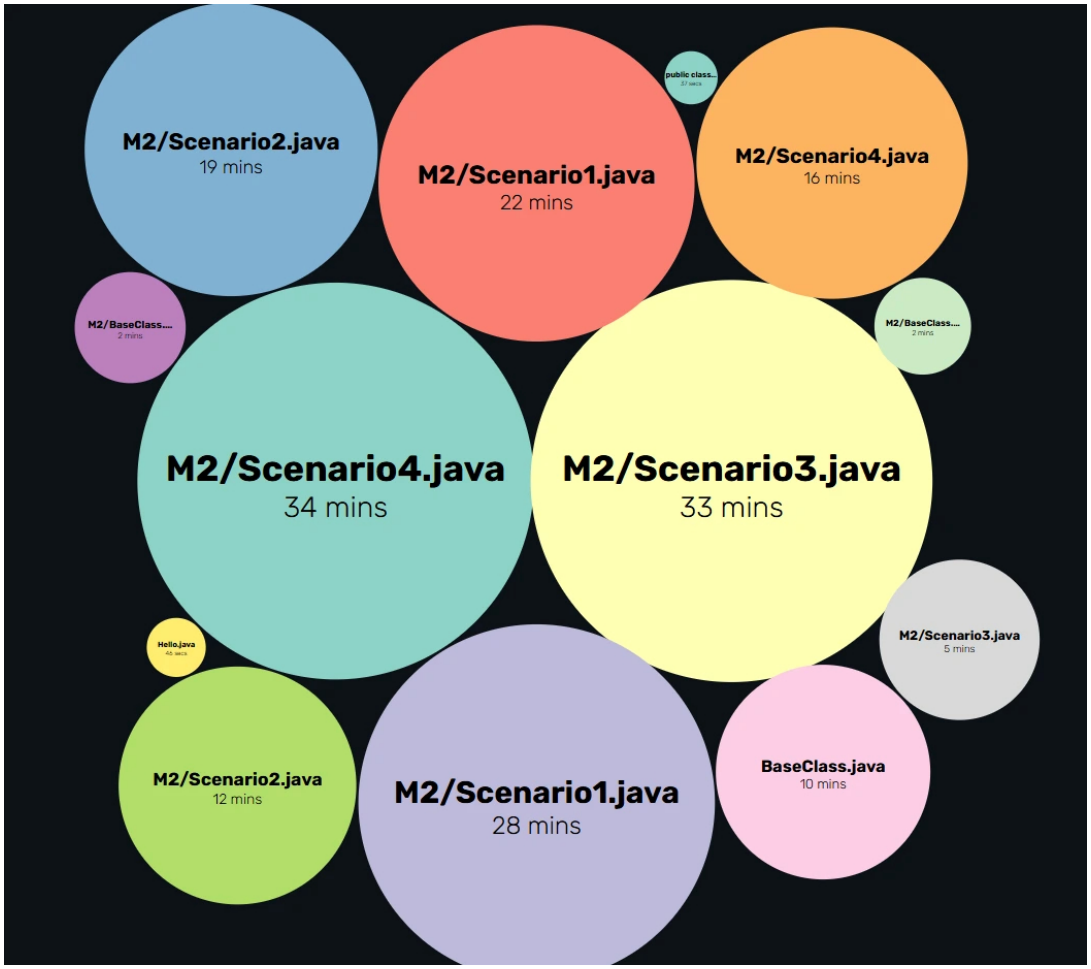
Image evidence prompt

Upload one readable WakaTime Projects image showing overall activity for your personal IT114 repository during the period when you worked on `M2-Homework`.

Requirements and potential deductions

- The image identifies the personal repository and shows overall assignment activity without exposing an API key; duration and graphs are not graded. (up to -0.1 points)

Caption: First Image



WakaTime file activity

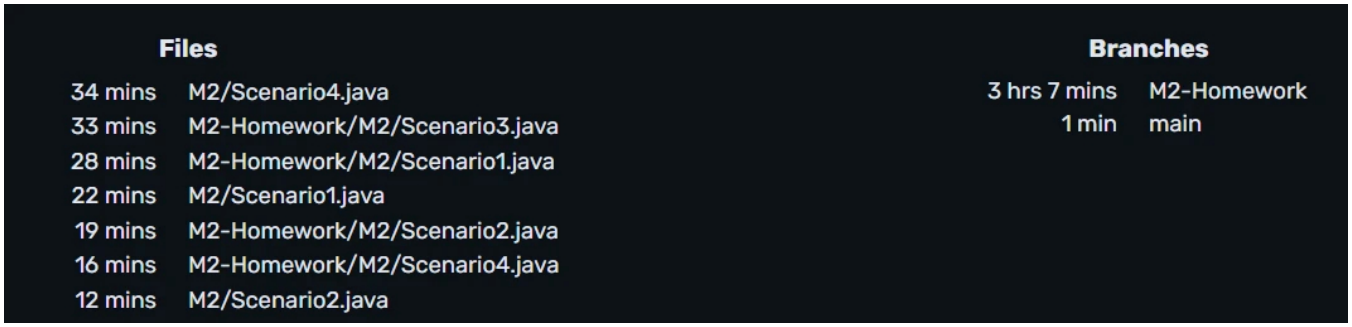
Image evidence prompt

Upload one readable WakaTime image showing Java file activity during the period when you worked on `M2-Homework`.

Requirements and potential deductions

- The image shows relevant Java file activity without exposing an API key; duration and graphs are not graded. (up to -0.1 points)

Caption: Second Image



Learned or verified

Response prompt

What did you learn or verify while completing the work on `M2-Homework`?

Requirements and potential deductions

- The response identifies something learned or verified through specific assignment experience. (up to -0.2 points)

Submitted response

I learned that if code works on the first compile then double check it. Also check if the output is exactly the required output and slightly different. Basically just follow the instructions and quadruple check the code.

Easiest part

Response prompt

What was easiest or most straightforward in the work on `M2-Homework`, and why?

Requirements and potential deductions

- The response identifies the easiest or most straightforward part with a concrete reason. (up to -0.2 points)

Submitted response

Making if and else statements or the like. I like conditional statements as it is easy to identify some is wrong by checking the condition.

Most difficult or uncertain part

Response prompt

What was most difficult or uncertain in the work on `M2-Homework`, or still needs work?

Requirements and potential deductions

- The response identifies a concrete difficulty, uncertainty, or remaining next step. (up to -0.2 points)

Submitted response

Small bugs like missing spaces or not quite right spelling is the bane of my coding. I find the bugs granted but I wasted about 1 minutes before realizing that I misspelled a variable when referencing or missed a space within Scenario 4 remove section.

Optional Extra Credit

Scenario 4 challenge 4 is optional and can add up to one extra-credit point. Submit its source, output, and explanation below if you attempt it; the ordinary maximum remains ten points.

Scenario 4 middle-character source

Extra-credit task

Complete optional challenge 4 in `transformText`: assign up to three middle characters to `placeholderForMiddleCharacters`, excluding the first and last characters for shorter phrases, or assign `Not enough characters` when appropriate.

URL prompt

Only if you attempted extra credit, submit the same `M2/Scenario4.java` GitHub URL you provided for Scenario 4 source, on `M2-Homework`, now including the middle-character solution.

Extra credit

Requirements and potential deductions

- The linked source assigns the requested middle-character extraction to `placeholderForMiddleCharacters` when characters remain. (up to -0.3 points)
- The linked source assigns `Not enough characters` when the phrase is too short for the extraction. (up to -0.3 points)

https://github.com/aia47-cyber/aia47-IT114-001-f2026/blob/M2-Homework/M2/Scenario4.java

Scenario 4 middle-character output

Extra-credit task

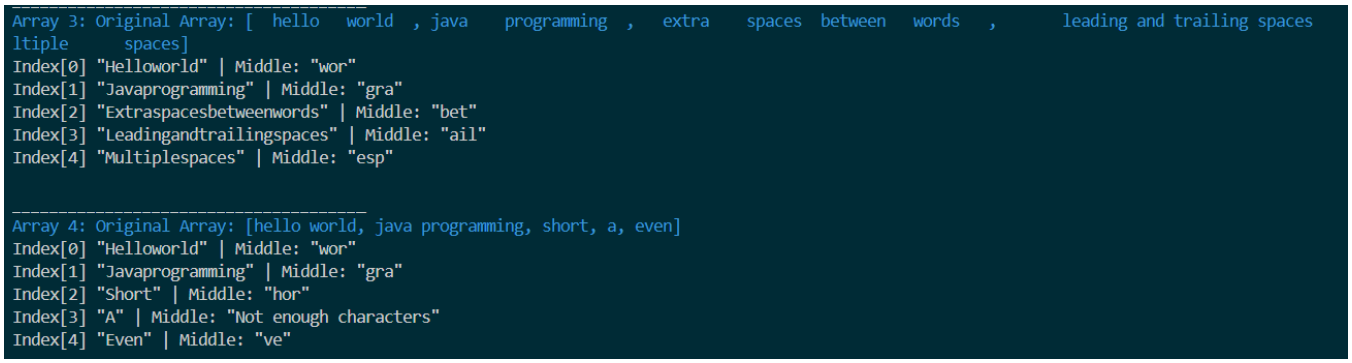
Image evidence prompt

Optionally upload one readable image of the optional `Scenario4` output produced from `M2-Homework`.

Extra credit

Requirements and potential deductions

- The output shows the requested middle-character result for the applicable starter inputs. (up to -0.1 points)
- The output shows `Not enough characters` for the too-short starter input. (up to -0.1 points)



Scenario 4 middle-character explanation

Extra-credit task

Response prompt

Optionally explain how the `M2-Homework` code chooses the middle characters and handles a value that is too short.

Extra credit

Requirements and potential deductions

- The response explains the extraction decision used in the linked code. (up to -0.1 points)
- The response explains when the linked code uses the too-short fallback. (up to -0.1 points)

Submitted response

No response submitted.